

DriftSense: Participant-Calibrated Early Prediction and Reflective Support for Browser Task Drift

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Fixed timers and domain-based controls cannot distinguish normal task behavior from activity that is unusual for a particular task or participant. We present DriftSense, a privacy-preserving Chrome extension and local machine-learning pipeline for studying browser task drift, defined as participant-reported deviation from a task declared at session start. Participants initiate a session, select a structured task type and intended duration, and use participant-approved task sites, including mixed-use sites when relevant. DriftSense records only approved hostname context, timing, aggregate interaction counts, focus and away-state summaries, and short activity windows; it does not collect page content, titles, messages, destination sites outside the approved set, or keystroke values. We specify a two-phase field protocol with 19–25 adult Chrome users. Phase 1 targets approximately 700 recorded sessions to obtain about 600 usable binary-labeled sessions over 10–14 days, with no model-assisted mid-session prompt. Models are evaluated at 3, 5, and 10 minutes, including a participant-calibrated logistic model whose relative features use prior sessions only. One checkpoint policy, model, threshold, and prompt cap are then frozen. During a seven- to ten-day Phase 2, model-eligible sessions are randomized equally to a nonblocking reflective side prompt or silent control. Prediction and intervention effects are evaluated separately. Empirical collection under this revised protocol has not occurred, so the manuscript reports the system and methodology without invented results. DriftSense does not claim to detect attention, objective productivity, emotion, addiction, or mental health status.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Ubiquitous and mobile computing systems and tools*; • **Computing methodologies** → *Machine learning approaches*.

Additional Key Words and Phrases: digital wellbeing, browser extension, task alignment, digital drift, self-report, privacy-preserving sensing, participant calibration, early prediction, reflective check-ins

1 Introduction

People routinely use the same browser sites for different tasks. YouTube may support a course, GitHub may support coding, Reddit may contain a technical answer, and a document editor may be open during focused writing or unrelated switching. Calling a hostname productive or distracting cannot establish whether the current activity remains aligned with the user’s task. Fixed elapsed-time rules are similarly limited because normal interaction patterns differ across writing, coding, research, learning, and communication.

Existing tools often operationalize undesirable use through screen time, app or website category, or fixed blocking rules [3, 13–15]. Such mechanisms can be useful for self-control, but they can also generate false positives when a user has a valid reason to visit a monitored website. They may also generate false negatives when a short session is clearly misaligned with intention. Recent work has therefore shifted toward user agency, reflection, friction, and intention alignment. For example, self-nudging systems such as *one sec* introduce a short pause before app access [9]; Purpose Mode reduces distracting interface patterns while preserving access to social media websites [11]; and PauseNow frames digital wellbeing as alignment between intentions and behavior [17].

We use the term *browser task drift* to describe participant-reported deviation from a structured task declared at session start. The label comes only from the participant’s post-session judgment; tab switching, inactivity, scrolling, leaving a task site, or a model score never defines drift. This framing is deliberately narrower than attention detection or productivity optimization. DriftSense studies task alignment without inferring clinical status, mental state, or semantic relevance of page activity.

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DriftSense uses a local-first Chrome Manifest V3 extension and a reproducible machine-learning pipeline. Participants choose sites they use for planned browser tasks; these *task sites* may include mixed-use sites and are not labeled productive or distracting. A participant initiates or confirms a task session and selects one structured type: writing or creating, coding or problem solving, reading or research, learning or tutorial, communication or coordination, or another planned task. The extension records content-free activity windows and creates leakage-safe snapshots at 3, 5, and 10 minutes. Participant-relative features compare the current pattern with prior aligned sessions without training a separate high-capacity model for each person. At the real session boundary, the participant reports whether the session remained aligned. In Phase 2, the frozen checkpoint policy determines eligibility for 1:1 prompt-versus-silent randomization, and access remains available in either assignment.

The central research questions are:

RQ1: At 3, 5, and 10 minutes, can structured task context and lightweight browser activity predict later self-reported task drift better than elapsed-time, intended-duration, task-site, task-only, and activity-only alternatives?

RQ2: Do participant-relative features derived only from prior sessions improve early prediction over an otherwise equivalent population model?

RQ3: Among sessions eligible under one frozen model policy, does a nonblocking reflective side prompt reduce subsequent self-reported task drift compared with a randomized silent control?

The expected contribution is an implementation and evaluation framework for participant-calibrated early task-drift prediction and lightweight reflective support. The contribution is not identifying productive websites, tracking activity, dashboarding, or blocking alone. It combines a user-declared task, privacy-minimized activity, participant-relative features derived without future leakage, explicit comparisons against simple policies, and a randomized test of one model-triggered prompt. Phase 1 supplies model-development data; it is not a causal control for Phase 2.

2 Related Work

2.1 Digital Self-Control Interventions

Digital self-control tools commonly use blocking, time limits, monitoring, reminders, friction, or interface modification [3, 13, 14]. Reviews find promising short-term effects but also short deployments, nonstandard outcomes, homogeneous samples, and few control groups; time spent is common but too coarse to represent why a site was used [14, 15]. Purpose Mode instead lets users suppress attention-capturing interface patterns while preserving access and found that goal alignment was more informative than site context alone [11]. These findings justify retaining time and domain as baselines rather than treating them as ground truth.

2.2 Intention and Reflection

Several systems create a reflective moment around technology use. *one sec* adds pre-access friction and an explicit choice to continue [9]; PauseNow proposes intention alignment but reports no outcome evaluation [17]; and StayFocused and MindShift deliver reflective or generated support, with mixed behavioral effects and concerns about repetitive prompts, assumptions, and richer personal data [12, 16]. WellScreen links estimated and actual smartphone use to daily reflection, but its small two-week deployment had no control condition [2]. DriftSense uses a shorter, nonblocking check-in and does not collect mental state.

Table 1. Closest systems informing the DriftSense research gap.

Work	Timing and mechanism	Role of intention	Evaluation	Remaining gap
Purpose Mode [11]	During-use interface modification	Goal alignment measured in EMA	Two-week field deployment	No session prediction; fixed phase order
<i>one sec</i> [9]	Friction before app access	Opening reason elicited around nudge	Observational field study and online experiment	No within-session prediction or field control
PauseNow [17]	Proposed recognition and nudging	Original intention is central	Conceptual framework	No deployed outcome evaluation
<i>Before You Scroll Again</i> [1]	Prediction before social-media use	Mismatch reported after use	Seven-day sensing study; within-person and held-out tests	Regret label; no intervention test
WellScreen [2]	Daily post-use reflection	Estimated versus actual use	Two-week mixed-methods deployment	No session classifier or control
DriftSense (planned)	Frozen early model plus one side prompt	Structured task declared before and revisited after use	10–14-day collection plus randomized 7–10-day intervention	Browser-only participant-calibrated pilot

2.3 Session-Level Modeling and Research Gap

Regret depends on activity and intention, not only duration [7, 10]. In a seven-day study with 21 participants, perceived intention–use mismatch correlated more strongly with regret than duration; phone-context CatBoost reached AUC 0.732 within person but only 0.536 when a participant was held out [1]. This gap motivates separate known-user and unseen-user evaluation. Behavioral sensing also creates privacy risks, so DriftSense limits collection to hostname-level context and aggregate interaction signals and makes no clinical or mental-state inference [4, 5].

Table 1 shows the closest comparison points. No evaluated system combines a declared pre-session browser intention, content-free early activity, participant-relative calibration, a post-session alignment label, explicit weak baselines, and randomized model-triggered reflection.

The contribution is therefore narrow: evaluate whether task context plus participant-relative, content-free activity improves early prediction over time, intended duration, task-site, task-only, and population-activity alternatives, then test the resulting prompt against a randomized silent control. The short pilot cannot establish durable behavior change, objective productivity, or population-level generalization.

3 System Design

3.1 Evidence-Derived Design Requirements

DriftSense is designed for a research setting where participants voluntarily install a browser extension, configure participant-approved task sites, initiate task sessions, and export a local dataset for analysis. Six requirements follow from the reviewed evidence:

- (1) **Measure task alignment, not assumed productivity.** Time and hostname remain baselines, but the target label must come from the participant’s declared task and later reflection. A mixed-use site may be task relevant, and leaving a task site may be necessary.

- (2) **Initiate an explicit task.** Merely opening a hostname is insufficient. The participant initiates or confirms a task session, selects a structured task type, and provides an intended duration as context rather than an automatic endpoint.
- (3) **Intervene lightly and preserve agency.** Reflection should provide an explicit choice and never block the site. A single neutral check-in follows evidence that friction and an opt-out can disrupt automatic action, whereas repetitive or inflexible prompts can be irritating [9, 12].
- (4) **Minimize and localize sensing.** The system uses aggregate interaction counts and approved-hostname context while excluding content, titles, screenshots, messages, key values, and destinations outside the approved task-site set. Outside-set activity is represented only as aggregate away time or state.
- (5) **Keep measurement stable across conditions.** The same task declaration and post-session questions should be used throughout collection and intervention so that the mid-session prompt is the only randomized interface difference.
- (6) **Make prediction and intervention auditable.** Exported records support 3-, 5-, and 10-minute prediction, participant-aware validation, participant-relative ablation, a frozen deployment policy, and verification of every eligibility and randomization decision.

3.2 Chrome Extension Architecture

The existing Chrome Manifest V3 prototype uses React and TypeScript, with a background service worker, content scripts, popup, options page, local dashboard, and allowlisted exports. It currently implements an earlier monitored-visit flow. Before the revised study it must be changed to support explicit task initiation, structured task types, multi-site task continuity, aggregate away-state sensing, recoverable pending reflection, and 3/5/10-minute snapshots. The Phase 2 build then adds a fixed local model, frozen checkpoint policy, 1:1 session-level randomization, and one nonblocking side prompt. No remote inference service is required. Earlier architecture and interface figures are withheld because they describe the superseded protocol.

3.3 Session Flow, Sensing, and Labeling

Participants select hostnames they use for planned browser tasks, including mixed-use sites when relevant. A task session begins only when the participant initiates or explicitly confirms it and selects writing/creating, coding/problem solving, reading/research, learning/tutorial, communication/coordination, or another planned task, together with an intended duration. The duration is context for modeling and reflection, not an automatic session end.

A task may continue across approved task sites. Leaving one approved site does not end the task or define drift. Outside the approved set, DriftSense stores only aggregate away time/state and never the destination. A session ends through explicit completion or a prespecified tested boundary such as confirmed abandonment or sustained inactivity. Closing or navigating away creates a recoverable pending reflection rather than silently discarding the label. Linked 10-second records store only offset, aggregate event counts, idle/focus/away state, and accessible video state. A separate intervention log records the frozen model, checkpoint, threshold decision, eligibility, assignment, prompt delivery/response, and suppression reason.

After each session, both study phases and both randomized assignments use the same question:

Did this session remain aligned with the task you started?

The primary binary labels are defined only by explicit *Aligned* and *No, I moved away from it* responses:

$$y = \begin{cases} 0, & \text{if answer is } \textit{Aligned} \\ 1, & \text{if answer is } \textit{No, I moved away from it} \end{cases} \quad (1)$$

Not sure and missing answers remain unlabeled and are excluded from the primary binary analyses. Outcome measurement is separate from actions such as continue, finish, remind later, or dismiss; an action never supplies the label.

During Phase 2, eligible sessions assigned to the prompt arm receive a compact side card asking *Still working toward your original goal?*; silent-control sessions receive no mid-session interface. The probability is not displayed, access remains available, and at most one prompt appears per session. The local dashboard supports participant control rather than forming a claimed contribution.

4 Drift Prediction and Model-Assisted Reflection

4.1 Prediction Task

The main task is binary session-level prediction:

Given structured task context, approved-hostname context, prior-session calibration, and lightweight activity observed by a fixed cutoff, predict whether the participant will later report that the task session drifted.

The positive class is the self-reported *No, I moved away from it* label and the negative class is *Aligned*. *Not sure* and missing responses are excluded from the binary task and reported separately. The model outputs a probability for audit and eligibility; it is not treated as attention detection or a fact about the session.

4.2 Feature Groups

Features cover task type, intended duration, approved task-site context, elapsed time, cumulative and recent activity, focus loss, tab switching, idle/active/away/video ratios, and event rates per active minute. Participant-relative features compare the current pattern with prior aligned sessions for the same or similar task type; all history is computed from sessions preceding the prediction. Activity windows are truncated at 3, 5, and 10 minutes. No early feature uses final duration, a later label, or later activity. Revised architecture and study-design figures will be generated only after the implementation and protocol are frozen.

4.3 Baselines

DriftSense evaluates simple policies and feature ablations before the participant-calibrated model:

- (1) **Majority baseline.** Predict the training-set majority class.
- (2) **Elapsed-time baseline.** Use a validation-derived fixed time policy.
- (3) **Intended-duration baseline.** Use elapsed time relative to intended duration.
- (4) **Task-site baseline.** Estimate approved-hostname tendency from training data only.
- (5) **Task-only logistic regression.** Use structured task type and intended duration without activity.
- (6) **Activity-only logistic regression.** Use population activity without task context.
- (7) **Task-plus-activity logistic regression.** Combine task context with population activity but omit participant-relative features.

These comparisons test both whether activity adds value beyond timer and task context and whether participant-relative representation adds value beyond the population model.

4.4 Main Models

The primary deployment candidate is logistic regression combining task context, activity, and participant-relative features. It is interpretable, appropriate for approximately 600 usable sessions, and straightforward to deploy locally. This is a participant-calibrated population model, not a separately trained model for every person. Random Forest is an optional robustness comparison, while CatBoost or XGBoost may appear only as secondary offline checks [6, 8]. GRU, TCN, Transformer, and high-capacity per-person models are outside this pilot because 19–25 participants and roughly 24–32 usable sessions per person on average do not justify them.

4.5 Early Prediction

Offline evaluation compares 3-, 5-, and 10-minute cutoffs. For each cutoff, reporting includes the number of sessions still observable, complete, and binary labeled. Results are shown both for each cutoff-specific cohort and for the common subset lasting at least 10 minutes, since later cutoffs otherwise select a different population. Model selection considers discrimination, calibration, coverage, expected prompt rate, and false-prompt burden. Before Phase 2, the study freezes either one selected checkpoint or a prespecified first-threshold-crossing policy across the three checkpoints, together with preprocessing, threshold, and prompt cap. At most one prompt can occur per session.

5 Methodology

5.1 Study Design

We plan a two-phase in-the-wild feasibility study with one cohort, preceded by a small technical and usability pilot. Phase 1 is a 10–14-day observational collection period with task initiation and post-session reflection but no model-assisted mid-session prompt. After model development, Phase 2 is a seven- to ten-day intervention period using the frozen policy. Sessions crossing the frozen threshold are randomized 1:1 to a reflective prompt or silent control. This randomized comparison addresses RQ3; Phase 1 prediction addresses RQ1 and RQ2. Overall before–after change is descriptive because phase order, novelty, and temporal changes prevent causal attribution.

The target is 19–25 completing participants. Phase 1 aims for approximately 700 recorded sessions to obtain about 600 usable binary-labeled sessions after missing responses, short sessions, and incomplete windows, with at least 20 usable sessions per retained participant and preferably 30–40. Phase 2 aims for 150–200 randomized eligible sessions where feasible, with most participants represented in both assignments. These are operational feasibility targets rather than a formal power calculation, and clustered sessions do not replace independent participants. Shortfalls will narrow the claims and keep intervention estimates exploratory.

5.2 Participants and Recruitment

Eligible participants are adults who use Chrome or a Chromium-based browser on a personal computer, regularly perform planned browser tasks, can identify participant-approved task sites, and can install the extension for the study period. Recruitment may use a university convenience sample of students and knowledge workers. Participants are not screened on addiction, ADHD, mental health, or another clinical category. Reporting includes recruitment source,

demographics relevant to interpretation, attrition, usable sessions, cutoff coverage, and missing labels without claiming representativeness.

5.3 Consent, Privacy, and Data Handling

Before installation, participants receive a consent form and field-level explanation. Consent covers approved task-site hostnames, structured task type, intended duration, session and checkpoint timing, aggregate click/scroll/keyboard-activity counts without key values, idle/active time, focus loss, tab switches, aggregate time outside the task-site set without destination identity, activity windows, post-session answers, and—during Phase 2—model version, estimate, random assignment, prompt display, and response. The extension does not collect page text, titles, paths, queries, passwords, screenshots, private messages, source code, free-text task descriptions by default, keystroke values, full browsing history, outside-set destinations, webcam data, identity, emotion, or inferred mental state.

Participants receive anonymous codes such as P01. They can pause monitoring, change monitored domains, inspect records, export their data, delete individual sessions, or delete all local data. Participant exports are stored in a private, access-controlled study directory outside the source repository. Only consented research staff can access the combined dataset. Any publication artifact will contain aggregates or de-identified excerpts rather than raw participant files. Ethics approval or institutional exemption, as applicable, must be obtained before recruitment.

5.4 Phase 1: Observational Collection

After onboarding, each participant chooses sites used for planned tasks, including mixed-use sites when relevant. The participant initiates or confirms each task session, selects a structured task type and intended duration, and then browses normally while aggregate session and activity-window signals are stored locally. At the real session boundary, the extension asks whether the session remained aligned with the task. Navigation or closure must preserve a pending reflection rather than silently treating the session as non-drift. No mid-session risk estimate or model-assisted intervention is shown during Phase 1.

Participants use the collector for 10–14 days. The research team checks only adherence and technical failure; it cannot inspect page content because none is collected. Participants export the session and activity-window files through the consented transfer procedure. Files are validated and combined outside the repository and Git history.

5.5 Model Development and Freeze

Earlier Phase 1 sessions form the development period and a prespecified later portion forms the chronological holdout for known-participant future-session performance. The exact split rule is frozen before modeling. Preprocessing, participant histories, features, hyperparameters, and threshold rules are selected within development data using participant-grouped folds. The chronological holdout is evaluated once. Participant-grouped or leave-one-participant-out evaluation is reported separately as unseen-user performance and is not conflated with the deployment-relevant known-participant result.

The primary deployment candidate is task-plus-activity logistic regression with participant-relative features. It is compared with majority, elapsed-time, intended-duration, task-site, task-only, activity-only, and non-calibrated task-plus-activity alternatives; Random Forest is optional. Metrics are accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, calibration where feasible, cutoff coverage, and expected prompt rate. Performance is computed at 3, 5, and 10 minutes, and every cutoff feature uses only information observed by that time. Participant-relative features use prior sessions only.

After holdout evaluation, one checkpoint policy, model, preprocessing specification, threshold, and prompt cap are frozen. A single selected checkpoint is preferred; a first-threshold-crossing policy across 3, 5, and 10 minutes is permitted only if prespecified and still produces at most one prompt per session. The threshold balances sensitivity, calibration, coverage, false prompts, and burden and is never retuned from Phase 2 outcomes. If enrichment over simple baselines is negligible or eligibility is impractical, the result is reported honestly and Phase 2 is framed, at most, as technical feasibility.

5.6 Phase 2: Micro-Randomized Intervention

The same participants use the intervention build for seven to ten days. Task declaration, sensing, boundaries, and post-session reflection remain identical to Phase 1. A session becomes eligible only at the frozen checkpoint policy, with complete features, a score meeting the frozen threshold, no prior assignment, and no suppression by the prompt cap. The extension uses local random assignment with equal probability:

- (1) **Reflective-prompt assignment.** Show a compact, nonblocking side card asking *Still working toward your original goal?* with continue, finish, remind-later, and dismiss actions.
- (2) **Silent-control assignment.** Show no mid-session interface.

Assignment is logged before rendering and survives reload or service-worker suspension. Analysis follows assignment regardless of interaction. A maximum of one prompt per session and a prespecified daily cap limit burden. Sessions below threshold, ending before the applicable checkpoint, missing features, or suppressed by the cap remain available for descriptive reporting but are excluded from the randomized prompt-effect comparison. Both assignments receive the same post-session reflection. This comparison estimates the effect of showing the prompt among model-eligible sessions; it does not directly compare model targeting with fixed-time prompting.

5.7 Measures

The primary prediction outcomes are F1-score and ROC-AUC for participant-calibrated task-plus-activity prediction relative to the required baselines. Accuracy, precision, recall, confusion matrices, class balance, calibration where feasible, cutoff coverage, and expected prompt rate are also reported. Results separate chronological known-participant prediction from participant-held-out prediction.

The primary intervention outcome is the proportion of complete binary-labeled eligible sessions reported as drift, compared by randomized assignment. Secondary outcomes include duration after assignment, immediate ending, prompt response and frequency, label completion, delivery failure, suppression, perceived timing, usefulness, annoyance, ease of use, and willingness to continue. Phase 1 versus Phase 2 changes remain descriptive.

At the end of Phase 2, participants rate five statements on a five-point agreement scale: the prompts appeared at appropriate times; the prompts helped them reconsider their intention; the prompts were annoying; the extension was easy to use; and they would continue using it. One open-ended question asks what should change about prompt wording or timing. These measures assess acceptability and burden, not clinical benefit or objective attention.

5.8 Data Processing and Quality Checks

The preprocessing pipeline validates exact schemas, participant identifiers, duplicate sessions, invalid overlap, cutoff completeness, and non-binary outcomes. Original exports are never modified. Binary analyses include only explicit *Aligned* (0) and *No, I moved away from it* (1); *Not sure* and missing outcomes remain separate.

Session, task-site, activity-window, and intervention records are joined by `session_id`. Categorical encoders, scalars, participant histories, and aligned-session reference distributions are fitted within training data and applied unchanged to validation or test data. Quality reporting includes recruited/completed participants, recorded and usable sessions, sessions per participant, class distribution, label completion, cutoff coverage, missing windows, rejected files, intervention eligibility, assignment balance, suppression, delivery failure, and attrition.

5.9 Analysis Plan

For RQ1, performance is summarized across predefined splits with participant-resampled 95% confidence intervals where feasible. Comparisons ask whether task-plus-activity improves over time, duration, site, task-only, and activity-only alternatives. For RQ2, ablation compares the participant-calibrated model with the otherwise equivalent population model. A result is not interpreted as new-user generalization unless participant-held-out performance supports that claim.

For RQ3, analysis is restricted to randomized eligible sessions with binary post-session labels and follows assigned condition. For each participant represented in both assignments, drift proportions are calculated separately for prompt and silent control. The prespecified simple comparison is a paired participant-level estimate with a participant-bootstrap 95% confidence interval and a paired test where appropriate. Assignment-specific missing-label and delivery-failure rates are reported. If eligible counts support it, a mixed-effects logistic regression with participant random intercept and assignment as the main fixed effect is a sensitivity analysis.

Secondary outcomes are analyzed descriptively or with paired participant-level comparisons. Open-ended responses are summarized with a transparent lightweight content analysis. No individual mechanism is declared effective from a Phase 1–Phase 2 before–after difference alone. Results will be described as short-term and preliminary; a seven-day intervention cannot establish durable behavior change. No results, significance values, or effect sizes will be inserted until the real consented dataset has been analyzed.

5.10 Reproducibility and Reporting Commitments

Before recruitment, the study materials will specify task types, task-site setup, session boundaries, pending reflection, outcome wording, binary-label mapping, checkpoint selection, exclusions, and confirmatory/exploratory outcomes. Before Phase 2, the extension records a versioned model artifact, preprocessing specification, checkpoint policy, threshold, prompt cap, and randomization procedure. Reporting traces participants and sessions from recorded Phase 1 data through the RQ1/RQ2 sets and from evaluated Phase 2 sessions through the RQ3 set, including short sessions, missing labels/windows, threshold eligibility, cap suppression, assignment, and delivery failure.

Code, schemas, isolated non-participant test fixtures, and analysis scripts may be released after review, but raw participant exports will not be placed in the public repository. A de-identified derived dataset will be shared only if consent, ethics approval, and disclosure-risk review permit it. Otherwise, reproducibility will rely on field-level documentation, deterministic preprocessing, fixed random seeds, environment information, and aggregate or disclosure-safe outputs.

5.11 Anticipated Validity Boundaries

The primary construct is a participant’s post-session judgment of alignment, not an objective measurement of attention. Labels may vary across people and may be influenced by recall, social desirability, repeated prompting, or the intervention

itself. Reporting label-completion and non-binary outcomes by assignment is therefore part of the main validity assessment rather than a housekeeping detail.

The RQ3 causal estimate applies only to sessions meeting the frozen model’s eligibility rule and not suppressed by the cap. It estimates the effect of showing the prompt within that eligible set, not the value of model targeting versus fixed-time prompting and not the effect of prompting every task or user. Randomization protects the prompt-versus-silent comparison, while fixed phase order prevents causal interpretation of overall Phase 1–Phase 2 change. Six hundred sessions remain clustered within only 19–25 participants, participant-relative histories are sparse, 10-minute evaluation selects longer sessions, and hostname/activity signals cannot establish semantic task relevance. A seven- to ten-day intervention supports only short-term feasibility and preliminary efficacy.

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